Individual Assignment 5

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1. Setup

Packages

```
# general data wrangling and plotting
library(tidyverse)

# file paths
library(here)

# clean column names
library(janitor)

# convert taxon names to snake case
library(snakecase)

# wrap long axis labels
library(scales)

# read Excel files
library(readxl)

# combine plots
library(patchwork)
```

Data

```r
# read in taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# read in aquatic invertebrate data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# read in water quality data
# na values are included so unusual entries are treated as missing
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

2. Cleaning

NCOS sites

```r
# create object with only NCOS quarterly sampling sites
ncos_sites <- sites |>

  # keep sites in the NCOS sector that were sampled quarterly
  filter(sector == "NCOS" &
           sampling_frequency == "Quarterly")
```

Taxon list

```r
# clean taxon names so they match the aquatic invertebrate data
taxon_list_clean <- taxon_list |>

  # convert verbatim taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

Water quality

```
# create cleaned water quality data frame
water_quality_clean <- water_quality |>

  # clean column names
  clean_names() |>

  # keep only NCOS quarterly sampling sites
  semi_join(ncos_sites, by = "site") |>

  # create a shared date-site column for joining later
  unite("date_site",
        date,
        site,
        remove = TRUE) |>

  # group by each date and site combination
  group_by(date_site) |>

  # calculate median water quality values for each date-site survey
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

  # ungroup data frame
  ungroup() |>

  # remove known salinity outlier
  filter(date_site != "2022-08-12_NVBR")
```

Aquatic invertebrates

```
# create cleaned aquatic invertebrate data frame
aq_ins_clean <- aq_ins |>

  # clean column names
  clean_names() |>

  # keep only NCOS quarterly sampling sites
```

```r
semi_join(ncos_sites, by = "site") |>

# rename date column for easier use
rename(date = date_on_vial) |>

# keep date, sample type, site, and major invertebrate taxon columns
select(date,
       sample_type,
       site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>

# keep only filtered beaker samples
filter(sample_type %in% c("FB 250", "FB250")) |>

# convert taxon columns into long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>

# group by date, site, and taxon
group_by(date, site, taxon) |>

# calculate average abundance per liter
summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5,
          .groups = "drop") |>

# create shared date-site column for joining with water quality
unite("date_site",
      date,
      site,
      remove = FALSE) |>

# join water quality data
left_join(water_quality_clean, by = "date_site") |>
```

```
  # join taxonomic information
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

  # create full site names
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~ "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>

  # reorder sites by median salinity
  mutate(site_full = fct_reorder(site_full,
                               med_sal,
                               .fun = "median",
                               .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

3. Class figures needed for assignment

Site salinity

```
# create object from aq_ins_clean
salinity <- aq_ins_clean |>

  # remove rows only if median salinity is missing
  filter(!is.na(med_sal)) |>

  # create plot
  ggplot(mapping = aes(x = site_full,
                       y = med_sal)) +

  # show individual salinity measurements for each survey
  geom_jitter(height = 0,
              width = 0.1,
```

```r
              shape = 21) +

  # wrap long site names on x-axis
  scale_x_discrete(labels = label_wrap(width = 15)) +

  # show median salinity for each site
  stat_summary(fun = median,
               geom = "point",
               color = "red",
               shape = 18,
               size = 4) +

  # label axes and add title
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +

  # use theme different from default
  theme_bw() +

  # angle x-axis text for readability
  theme(axis.text.x = element_text(angle = 45,
                                   hjust = 1))
```

Copepod abundance

```r
# create data frame for copepod abundance
copepod_data <- aq_ins_clean |>

  # keep only copepods
  filter(taxon == "copepod") |>

  # log + 1 transform average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```r
# create copepod plot object
copepods <- ggplot(data = copepod_data,

                   # x-axis: site
                   # y-axis: log-transformed average abundance per liter
```

```r
                 mapping = aes(x = site_full,
                               y = log_ave_lit)) +

  # show individual average abundance/L measurements
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +

  # show median abundance for each site
  stat_summary(fun = median,
               geom = "point",
               color = "blue",
               shape = 18,
               size = 4) +

  # wrap long site names on x-axis
  scale_x_discrete(labels = label_wrap(width = 15)) +

  # label axes and add title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +

  # use theme different from default
  theme_bw() +

  # angle x-axis text for readability
  theme(axis.text.x = element_text(angle = 45,
                                   hjust = 1))
```

4. Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan the figure

Goal: create a plot showing differences between sites in ostracod abundance.

- x-axis: `site_full`
- y-axis: `log_ave_lit` (log + 1 transformed average abundance/L)
- geometry to represent average abundance/L: jittered points using `geom_jitter()`

- geometry to represent medians: larger colored diamond points using `stat_summary(fun = median)`

b. Wrangle data

```
# create data frame for ostracod abundance
ostracod_data <- aq_ins_clean |>

  # keep only ostracods
  filter(taxon == "ostracod") |>

  # log + 1 transform average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows from ostracod data frame
ostracod_data |>
  slice_sample(n = 10)
```

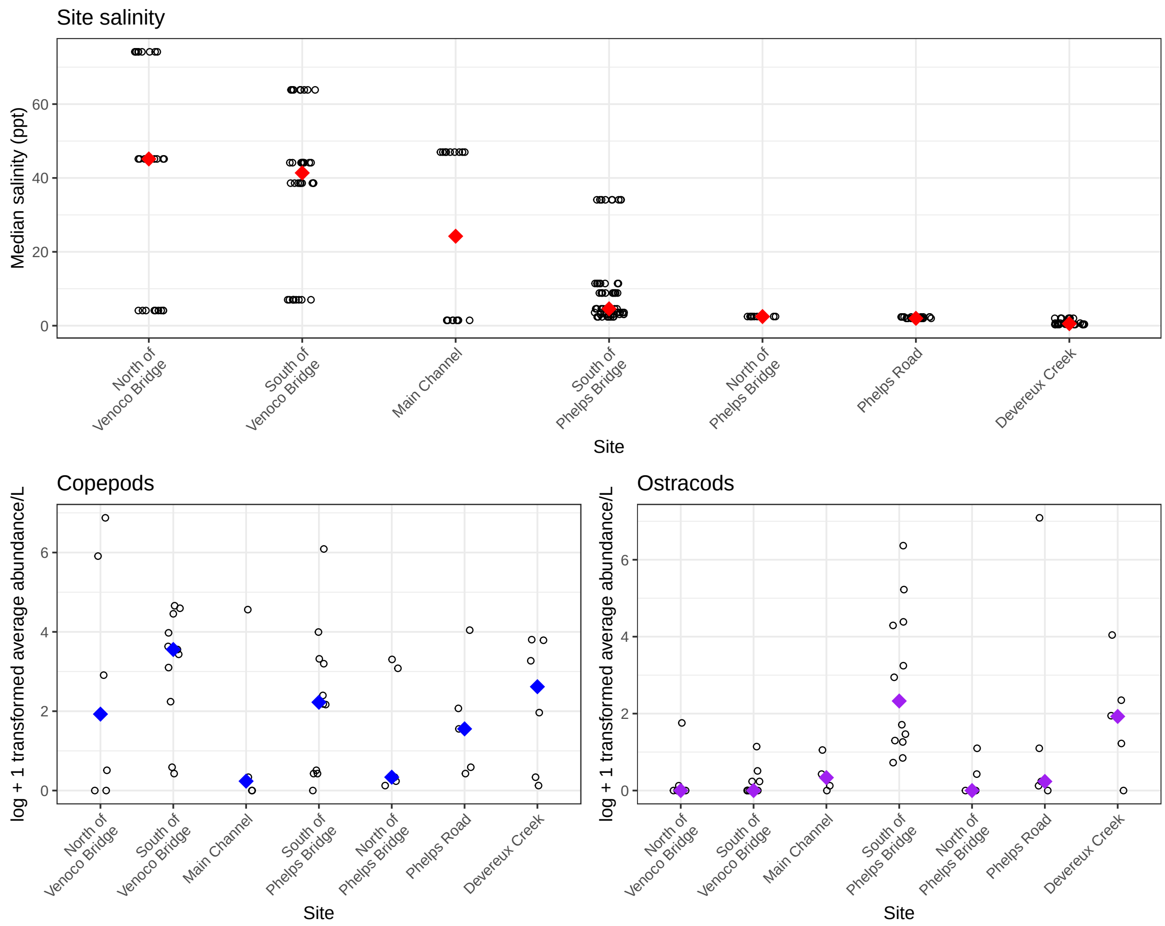
```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-08-12_NVBR 2022-08-12 00:00:00 NVBR  ostr~   0      NA      NA     NA
 2 2022-08-08_NPB2 2022-08-08 00:00:00 NPB2  ostr~   2       7.68    2.36   2.43
 3 2022-11-19_NDC  2022-11-19 00:00:00 NDC   ostr~   6      NA      NA     NA
 4 2022-05-13_NVBR 2022-05-13 00:00:00 NVBR  ostr~   0       9.72   74.2   11.4
 5 2023-09-15_NMC  2023-09-15 00:00:00 NMC   ostr~   0.4    NA      NA     14.2
 6 2023-05-04_NEC  2023-05-04 00:00:00 NEC   ostr~   0.667   8.05   NA      8.55
 7 2022-02-25_NEC  2022-02-25 00:00:00 NEC   ostr~   0       8.75   44.1    9.87
 8 2023-03-03_NPB2 2023-03-03 00:00:00 NPB2  ostr~   0.133   7.71    1.98   9.59
 9 2021-08-25_NPB  2021-08-25 00:00:00 NPB   ostr~ 581.      7.52    4.57   4.5
10 2023-11-11_NEC  2023-11-11 00:00:00 NEC   ostr~   0      NA      38.6    0.25
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code figure

```
# create ostracod plot object
ostracods <- ggplot(data = ostracod_data,

                    # x-axis: site
                    # y-axis: log-transformed average abundance per liter
                    mapping = aes(x = site_full,
                                  y = log_ave_lit)) +

  # show individual average abundance/L measurements
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +

  # show median abundance for each site with a visually distinct point
  stat_summary(fun = median,
               geom = "point",
               color = "purple",
               shape = 18,
               size = 4) +

  # wrap long site labels
  scale_x_discrete(labels = label_wrap(width = 15)) +

  # label axes and add title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +

  # use theme different from default
  theme_bw() +

  # angle x-axis text for readability
  theme(axis.text.x = element_text(angle = 45,
                                   hjust = 1))
```

d. Create multipanel figure

```
# combine salinity, copepod, and ostracod plots
salinity / (copepods | ostracods)
```


Site salinity
Median salinity (ppt)
0
20
40
60
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
Phelps Road
Devereux Creek
Site
Copepods
log + 1 transformed average abundance/L
0
2
4
6
Site
Ostracods
log + 1 transformed average abundance/L
0
2
4
6
Site